

Kazuki Fujii

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EDUCATION

Institute of Science Tokyo (formerly Tokyo Institute of Technology) Ph.D. in Computer Science – Advisor: Prof. Rio Yokota	April 2026 – March 2029 (expected)
Institute of Science Tokyo (formerly Tokyo Institute of Technology) M.Eng. in Computer Science	April 2024 – March 2026
Tokyo Institute of Technology B.Eng. in Computer Science	April 2020 – March 2024

EXPERIENCE

Upcoming NVIDIA Internship Apr 2027 – Sep 2027

PhD Research Intern – Efficient LLM Inference – NVIDIA
Santa Clara, CA, USA | Jun 29, 2026 – Sep 18, 2026

- **Agentic RL (SWE-RL) Workload Analysis:** Analyzed bottlenecks across prefill, decode, and tool-call phases in real SWE-RL workloads to identify opportunities for throughput improvement.
- **Concurrency and Resource Usage:** Analyzed the interplay between in-flight batch size (rollout concurrency), sandbox load, and KV cache eviction to understand system behavior under different workload settings.
- **Policy Staleness Metrics:** Proposed metrics to quantify and analyze policy staleness in asynchronous RL, enabling comparisons of staleness across training settings.
- **Throughput and Staleness:** Conducted analyses and experiments on the relationships among training settings, policy staleness, throughput, and the reduction in learning effectiveness per training step caused by staleness.

Research Intern – Swallow LLM Project – AIST
Tokyo, Japan | Oct 2023 – Present

- **Large-Scale Pre/Post-Training:** Core contributor to all medium-to-large training runs using Megatron-LM and NeMo frameworks, leading to the release of ten bilingual LLM series. First-generation models (Dec 2023) achieved SOTA among Japanese LLMs (201 citations).
- **Dataset Refinement (SwallowCode & SwallowMath):** Led research planning and data construction for v1 (May 2025) and v2 (Nov 2025). SwallowCode outperformed Stack-Edu by +17.0 pass@1 on HumanEval; SwallowMath surpassed finemath-4+ by +12.4 on GSM8K.
- **Low Precision (FP8) Training Research:** Validated Blockwise FP8 (E4M3) for 70B-scale models, demonstrating 1.18x FLOP/s/GPU speedup with no accuracy degradation. Showed that Delayed Scaling and per-tensor Current Scaling cause instability at scale, challenging prior assumptions.
- **Open Source:** Authored llm-recipes (44★) and moe-recipes (21★). Collaborate with NVIDIA’s Megatron-LM team (PM: Santosh Bhavani) on roadmap alignment.

Research Intern / Research Engineer – Turing Inc.
Tokyo, Japan | Feb 2023 – Jun 2026

- Research Intern (Feb 2023 – Mar 2026); Research Engineer (Apr 2026 – Jun 2026).
- **VLM and VLA model training infrastructure:** Co-authored vlm-recipes (20★), a PyTorch FSDP-based distributed training library, adopted as the core training library for the Heron VLM series.
 - **Cluster management:** Collaborated with Google Cloud engineers to deploy and maintain a 256-GPU cluster; delivered a joint talk at Google Cloud Next’24 Tokyo.
 - Authored technical blog posts on LLM training and infrastructure (1600+ likes), boosting the company’s engineering visibility.

Research Engineer Intern – Preferred Networks
Tokyo, Japan | Jun 2025 – Sep 2025

- Improved trillion-token pre-training data for PlaMo LLM by applying SwallowCode refinement techniques, achieving 10+ point improvement on HumanEval+.
- Validated FP8 (E4M3) blockwise training stability via continued pre-training from Llama-3.1-8B and Qwen3-8B over hundreds of billions of tokens, confirming 1.18x speedup without downstream task performance degradation.
- Investigated BPC as a proxy metric for benchmark scores (HumanEval, MMLU); findings published as an official company technical blog post.

SELECTED PUBLICATIONS

Peer-Reviewed Conference Papers

- **Rewriting Pre-Training Data Boosts LLM Performance in Math and Code** (ICLR2026) – 28 citations – First Author
Proposed LLM Rewriting (Transform-and-Retain): synthesizes high-quality pre-training data by upgrading low-quality samples rather than discarding them. Led the full project lifecycle from inception to dataset release. SwallowCode outperformed Stack-Edu by +17.0 pass@1 (HumanEval); SwallowMath surpassed finemath-4+ by +12.4 (GSM8K). Released SwallowMath and SwallowCode v1 & v2. Methodology adopted and cited by **Nemotron-3** (NVIDIA), **OLMo-3** (AI2), and **Kimi K2** (Moonshot AI).
- **Continual Pre-Training for Cross-Lingual LLM Adaptation: Enhancing Japanese Language Capabilities** (COLM2024) – 201 citations – First Author
Built Swallow LLM via continual pre-training of Llama 2, achieving SOTA among open Japanese LLMs (Dec 2023). Identified JP/EN data mixing (9:1) to preserve English performance; showed vocabulary expansion improved generation efficiency by 78% with a summarization task performance trade-off.
- **Drop-Upcycling: Training Sparse Mixture of Experts with Partial Re-initialization** (ICLR2025) – 23 citations – Co-Author
Novel upcycling method converting dense LLMs to sparse MoE by partially re-initializing FFN experts, improving training efficiency over conventional full upcycling.
- **Building a Large Japanese Web Corpus for Large Language Models** (COLM2024) – 49 citations – Co-Author
Constructed a large-scale Japanese web corpus via domain-aware filtering and deduplication pipelines, serving as the data foundation for Swallow LLM series.

Workshop Papers

- **Ilm-recipes: A Framework for Seamless Integration and Efficient Continual Pre-Training of LLMs** (SC24 TPC) – First Author
Led development of a custom training framework (PyTorch FSDP v1) for 0-day support of new LLMs not yet supported by Megatron-LM, enabling SFT and continual pre-training for any HuggingFace compatible models.

Preprints

- **Accelerating Large Language Model Training with 4D Parallelism and Memory Consumption Estimator** (arXiv 2024) – 6 citations – First Author
Formulated a precise per-GPU memory equation for 4D parallelism (TP/CP/DP/PP) covering model states, optimizer states, and activations. Validated via 454 large-scale experiments.
- **Balancing Speed and Stability: The Trade-offs of FP8 vs. BF16 Training in LLMs** (arXiv 2024) – 3 citations – First Author
Investigated FP8 stability for continual pre-training of 70B models. Discovered that DelayedScaling (TransformerEngine v1.x) is insufficient for this regime, causing instability in training from Llama-3-70B.

SELECTED INVITED TALKS

2026

- **NVIDIA GTC 2026** (Session S81710) – Swallow LLM: Continual Pre-Training and RL for Sovereign AI
Covered Qwen3-Swallow/GPT-OSS-Swallow families, blockwise FP8 training, and DAPO based RLVR.

2025

- **DDN User Group Meeting Tokyo** – Storage and I/O optimization for large-scale LLM training workloads.
- **IHPCSS2025** (Lisbon) – Poster presentation at the International HPC Summer School on large-scale LLM training.
- **AWS Summit Japan 2025** (Session CUS-02) – Building the Japanese LLM “Swallow” using Amazon SageMaker HyperPod

2024

- **SC24** (Tokyo Tech Booth) – Continual Pre-Training on TSUBAME for a Target Language: methodology for adapting Llama-3 (8B/70B) on TSUBAME supercomputer.
- **NVIDIA AI Summit Japan 2024** – Insights on Training Large-Scale Models.
- **Google Cloud Next 24 Tokyo** – Accelerating Training with Google Cloud AI Hypercomputer: joint talk on GENIAC H100 (A3) cluster deployment with Google Cluster Toolkit.
- **RIKEN AIP Seminar** – Technical deep-dive on 3D Parallelism (TP/DP/PP) for efficient LLM training.

AWARDS & SERVICE

- 2024 **Best Paper Award**, IPSJ National Convention (Japan) – Distributed Training of LLMs
- 2024 **Best Paper Award**, Association for Natural Language Processing (Japan) – Continual Pre-Training of LLMs
- **Reviewer**: ICLR2026
- **Technical Writing**: 26 articles on Zen (Japan’s leading engineering blog platform), 1600+ likes; contributor for NVIDIA Technical Blog and AWS Technical Blog.